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BLOCKER DOOR EQUIPPED WITH LARGE CAVITY ACOUSTIC STRUCTURE

Abstract

The present disclosure provides a blocker door for a thrust reversing system of an aircraft engine assembly. The blocker door has a door body having a backplate. The blocker door also includes a perforated facesheet coupled with the door body. A plurality of ribs arranged between the backplate and the facesheet define a plurality of cavities that communicate with the plurality of perforations to create a plurality of acoustic cells.

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Background/Summary

FIELD
[0001] Aspects of the present disclosure generally relate to aircraft noise attenuation, and, more

particularly, to noise attenuating structures in an aircraft engine nacelle.

BACKGROUND

[0002] Ultra-high bypass ratio engines can provide significant fuel efficiency improvement but typically require larger nacelles to accommodate their increased fan diameters, which induces weight and drag penalties on aircraft. Compact architectures for these larger nacelles are being developed to help offset these inefficiencies. However, as nacelle architectures become more compact, the treatable acoustic area throughout the propulsion flow path is reduced. Moreover, some newer fans generate elevated tonal noise at certain frequencies. Smaller treatable acoustic areas and elevated tonal noise have presented certain challenges for aircraft engine assemblies to meet new, more stringent noise standards.

SUMMARY

[0003] The present disclosure provides a blocker door in one aspect. The blocker door includes a door body having a backplate. The blocker door also has a facesheet coupled with the door body and defining a plurality of perforations. Further, the blocker door has a plurality of ribs arranged between the backplate and the facesheet to define a plurality of cavities that communicate with the plurality of perforations to create a plurality of acoustic cells.

[0004] In one aspect, in combination with any example blocker door above or below, each cavity of the plurality of cavities has a length that is at least greater than ⅜ of an inch.

[0005] In one aspect, in combination with any example blocker door above or below, the blocker door defines a midline that separates the blocker door into a first side and a second side, and wherein the plurality of ribs includes a central-aft rib extending along the midline, a first-side aft rib angled with respect to the midline, a second-side aft rib angled with respect to the midline, a first-side central rib extending perpendicular to the midline, a second-side central rib extending perpendicular to the midline, a first-side central diagonal rib angled diagonal to the midline, and a second-side central diagonal rib angled diagonal to the midline, and wherein the central-aft rib, the first-side aft rib, the second-side aft rib, the first-side central rib, the second-side central rib, the first-side central diagonal rib, and the second-side central diagonal rib intersect at a central rib juncture.

[0006] In one aspect, in combination with any example blocker door above or below, the door body has an outer wall that extends from the backplate and along at least a portion of a perimeter of the door body, and wherein the central-aft rib, the first-side aft rib, the second-side aft rib, the first-side central rib, the second-side central rib, the first-side central diagonal rib, and the second-side central diagonal rib each respectively connect to the outer wall.

[0007] In one aspect, in combination with any example blocker door above or below, the door body has a hinge extending from the backplate in a direction opposite the plurality of ribs, and wherein the backplate defines a hinge recess in which the hinge is positioned, and wherein the door body includes a flange extending from the backplate, a mounting base, an outer wall, and wherein the plurality of ribs includes a partition rib and a hinge rib that extends between and connects the flange and the mounting base, and wherein the flange, the partition rib, the hinge rib, and the outer wall define a hinge cavity, which is one of the plurality of cavities, and wherein the hinge cavity mirrors the hinge recess along a thickness direction of the blocker door.

[0008] In one aspect, in combination with any example blocker door above or below, the hinge cavity has a portion that gradually increases in depth as the hinge cavity extends toward an end of the blocker door.

[0009] In one aspect, in combination with any example blocker door above or below, the door body includes a flange extending from the backplate and a mounting base that has one or more mounting structures, and wherein the plurality of ribs includes a first-side hinge rib and a second-side hinge rib that each extend between and connect the flange and the mounting base, and wherein the plurality of cavities includes a central forward cavity defined by the flange, the mounting base, the first-side hinge rib, and the second-side hinge rib.

[0010] In one aspect, in combination with any example blocker door above or below, the door body has an outer wall that extends from the backplate and along at least a portion of a perimeter of the door body, and wherein the door body includes a mounting base that includes one or more mounting structures, and wherein the plurality of ribs includes a central diagonal rib and a forward diagonal rib that each extend at an angle with respect to a midline of the blocker door and each extend perpendicular to one another, the central diagonal rib and the forward diagonal rib intersect at a rib juncture and each extend between and connect to the outer wall and the mounting base.
[0011] In one aspect, in combination with any example blocker door above or below, the blocker door has a forward end and an aft end and defines a midline that separates the blocker door into a first side and a second side, wherein the door body has an outer wall that extends from the backplate and along at least a portion of a perimeter of the door body, and wherein the door body includes a mounting base that includes one or more mounting structures, and wherein the plurality of ribs includes a first-side forward diagonal rib and a second-side forward diagonal rib, the first-side forward diagonal rib is arranged on the first side and extends between and connects to the outer wall and the mounting base, the second-side forward diagonal rib is arranged on the second side and extends between and connects to the outer wall and the mounting base, the first-side forward diagonal rib and the second-side forward diagonal rib converge toward one another as they extend toward the forward end.
[0012] In one aspect, in combination with any example blocker door above or below, the door body includes a mounting base that includes one or more mounting structures, and wherein the plurality of cavities includes a forward inner cavity defined by the mounting base, and a central diagonal rib and a forward diagonal rib of the plurality of ribs, and wherein the mounting base has a stepped sidewall that steps the mounting base inward toward a midline defined by the blocker door.
[0013] In one aspect, in combination with any example blocker door above or below, the blocker door defines a midline separating the blocker door into a first side and a second side, and wherein the plurality of ribs are arranged so that: at least two ribs of the plurality of ribs intersect at a first-side rib juncture on the first side of the blocker door, at least at least two ribs of the plurality of ribs intersect at a second-side rib juncture on the second side of the blocker door, and at least two ribs of the plurality of ribs intersect at a central rib juncture arranged coplanar with the midline.
[0014] In one aspect, in combination with any example blocker door above or below, an average volume of the acoustic cells ranges between 7 and 9 cubic inches.
[0015] In one aspect, in combination with any example blocker door above or below, the plurality of acoustic cells include ten to sixteen acoustic cells.
[0016] In one aspect, in combination with any example blocker door above or below, at least one of the plurality of acoustic cells is empty to allow for a volume of air to fill therein.
[0017] In one aspect, in combination with any example blocker door above or below, at least one of the plurality of acoustic cells has a volume of acoustic attenuating material disposed therein.
[0018] In one aspect, in combination with any example blocker door above or below, the ribs are integrally formed with the backplate as a monolithic unitary component.
[0019] The present disclosure further provides an aft fan duct assembly. The aft fan duct assembly includes an inner duct wall and an outer duct wall. The inner duct wall and the outer duct wall define a fan duct. The aft fan duct assembly also includes blocker doors arranged within the fan duct and each movable between a stowed position and a deployed position. Each one of the blocker doors includes a door body having a backplate and a plurality of ribs that extend from the backplate to define a plurality of cavities; and a facesheet coupled with the door body, wherein the facesheet encloses the plurality of cavities and defines a plurality of perforations that communicate with the plurality of cavities to create a plurality of acoustic cells.
[0020] In one aspect, in combination with any example aft fan duct assembly above or below, the plurality of ribs includes a set of ribs that intersect at a central rib juncture, wherein a first rib of the set extends along a midline defined by the blocker door, a second rib of the set extends

perpendicular to the midline, and a third rib of the set extends diagonally between the first and second rib.
[0021] In one aspect, in combination with any example aft fan duct assembly above or below, the blocker door defines a midline separating the blocker door into a first side and a second side, and wherein the plurality of ribs are arranged so that: at least two ribs of the plurality of ribs intersect at a first-side rib juncture on the first side of the blocker door, at least at least two ribs of the plurality of ribs intersect at a second-side rib juncture on the second side of the blocker door, and at least two ribs of the plurality of ribs intersect at a central rib juncture arranged coplanar with the midline.
[0022] The present disclosure further provides a method of fabricating a blocker door. The method includes creating a door body from a block of material, the door body having a backplate and a plurality of ribs that extend from the backplate to define a plurality of cavities. The method further includes coupling a facesheet to the door body to enclose the plurality of cavities and so that a plurality of perforations defined by the facesheet communicate with the plurality of cavities to create a plurality of acoustic cells.
[0023] The present disclosure provides a blocker door in yet another aspect. The blocker door includes a door body having a backplate and a plurality of ribs that extend from the backplate to define a plurality of cavities. The blocker door also includes a facesheet coupled with the door body. The facesheet encloses the plurality of cavities and defines a plurality of perforations that communicate with the plurality of cavities to create a plurality of acoustic cells.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] So that the manner in which the above recited features can be understood in detail, a more particular description, briefly summarized above, may be had by reference to example aspects, some of which are illustrated in the appended drawings.
[0025] FIG. **1** is a perspective view of an aft fan duct assembly for an aircraft engine assembly according to an example aspect of the present disclosure.
[0026] FIG. **2** is a view of a flowpath facing side of a blocker door according to an example aspect of the present disclosure.
[0027] FIG. **3** is a view of a non-flowpath facing side the blocker door of FIG. **2**.
[0028] FIG. **4** is an exploded view of the blocker door of FIG. **2**.
[0029] FIG. **5** is an assembled view of the blocker door of FIG. **2** and depicts acoustic cells thereof.
[0030] FIG. **6** is a perspective view of a side of a facesheet that confronts or faces a door body when coupled thereto, with the facesheet having a plurality of structurally-active ribs integrally formed therewith or coupled thereto, according to an example aspect of the present disclosure.
[0031] FIG. **7** is a flow diagram for a method of fabricating a blocker door according to an example aspect of the present disclosure.
DETAILED DESCRIPTION
[0032] Blocker doors can be incorporated into a thrust reversing system of an aircraft engine assembly and can be deployed to block or redirect fan bypass airflow after touchdown of an aircraft to ultimately produce reverse thrust, which aids in reducing the speed of the aircraft. When stowed during normal flight operations, blocker doors can form a boundary of a bypass flow duct. Disclosed herein is a blocker door equipped with features that can advantageously attenuate noise, such as aft-radiating noise from an aircraft engine. The acoustic features of the blocker door can reduce noise and mitigate elevated tonal noise of modern aircraft engines at certain frequencies, particularly during takeoffs and landings. Such features allow for aircraft engines to address new, more stringent noise standards and increased landing fees at airports. Moreover, the acoustic features can be integrated with or formed at least in part by structural features, providing for a

weight-efficient, structurally-sound, and acoustically-enhanced blocker door.

[0033] In some example aspects, a blocker door can include a door body having a backplate and a plurality of structurally-active ribs that extend from the backplate to define a plurality of cavities. The ribs are structurally-active in that they carry loads imparted to the blocker door. The ribs can be redesigned existing structural elements and can include additional “close-out” ribs or stiffeners to improve the impedance characteristics of the cavities. A relatively thin facesheet can be overlaid onto the door body and coupled thereto, e.g., with mechanical fasteners. The facesheet encloses the plurality of cavities but also defines a plurality of perforations that communicate with the plurality of cavities, which creates a plurality of acoustic cells or resonance chambers that can be used to attenuate community noise during takeoff, approach, landing, and cruise. The perforations in the facesheet enable fluid flow into the acoustic cells, allowing noise to be attenuated. In some alternative aspects, the structurally-active ribs can be integrally formed with the facesheet and coupled to the backplate when the facesheet is overlaid onto the door body. Each blocker door of a thrust reversing system can include such acoustic features. In this way, aft-radiating noise from an aircraft engine can be attenuated, which can provide improved acoustics in compact nacelle architectures, among other possible advantages, benefits, and/or technical effects.

[0034] As a further advantage, benefit, and/or technical effect, the blocker door of the present disclosure can enable the use of alternate materials and fabrication processes for acoustic attenuation capabilities, which can reduce costs over traditional methods. As yet a further advantage, benefit, and/or technical effect, the blocker door of the present disclosure can utilize sections of a door body that have not traditionally been utilized for acoustic attenuation purposes, such as proximate the hinge fittings. This can increase the acoustic attenuation capabilities of the blocker door. The blocker door of the present disclosure can have further advantages, benefits, and/or technical effects than those noted herein.

[0035] FIG. **1** is a perspective view of a portion of an aft fan duct assembly **100** according to an example aspect of the present disclosure. The aft fan duct assembly **100** can be incorporated into an aircraft engine assembly, for example. The aft fan duct assembly **100** can be located in an aft portion of the aircraft engine assembly, or more generally, an aft part of an aircraft nacelle. For reference, the aft fan duct assembly **100** defines an axial direction A, a radial direction R, and a circumferential direction C. The aft fan duct assembly **100** also defines a central axis CA, which extends along the axial direction A.

[0036] The aft fan duct assembly **100** has a forward end **102** and an aft end **104**. The aft end **104** is spaced from the forward end **102**, e.g., along the axial direction A. The aft fan duct assembly **100** has a bifurcated construction and includes a pair of barrel-shaped structures, or duct assemblies **106**, **108**. The duct assemblies **106**, **108** each include upper and lower fairings (or radial walls) that are coupled respectively with an upper beam (e.g., a hinge beam) and a lower beam (e.g., a latch beam). In this regard, the aft fan duct assembly **100** includes upper and lower bifurcation assemblies **110**, **112**. The duct assemblies **106**, **108** form an inner duct wall **114**. The inner duct wall **114** is spaced inward of an outer duct wall **116** along the radial direction R with respect to the central axis CA. A fan duct **118** (e.g., an ultra-high bypass fan duct), or rather a downstream or aft portion thereof, is defined between the inner duct wall **114** and the outer duct wall **116**. It will be appreciated that the aft fan duct assembly **100** can be coupled with, and positioned aft of, a forward fan duct assembly. The forward fan duct assembly can include a fan and can define an upstream portion of the fan duct **118**. A core turbine engine can extend through the forward fan duct assembly and the aft fan duct assembly.

[0037] The aft fan duct assembly **100** can include a thrust reversing system **120** or thrust reverser. The thrust reversing system **120** includes, among other things, a plurality of blocker doors **122** arranged within the fan duct **118**. The blocker doors **122** are circumferentially arranged about both bifurcated sections of the fan duct **118**. Each one of the blocker doors **122** can be movable between a stowed position and a deployed position. The blocker doors **122** can be individually or collectively actuated, e.g., by one or more actuators of the thrust reversing system **120**. The actuators can be arranged within the annulus between the outer duct wall **116** and an outer cowl **124** of the aft fan duct assembly **100**. Each one of the blocker doors **122** is coupled with the outer duct wall **116** via one or more hinge fittings and with the inner duct wall **114** via respective drag links **126**.

[0038] During normal flight operations, the blocker doors **122** can be moved to their respective stowed positions in which the blocker doors **122** are substantially retracted or removed from the fan duct **118** to allow for a fluid flow FF, e.g., a fan bypass air stream, to flow through the fan duct **118** and exit the aft end **104**. The blocker doors **122** are shown in their respective stowed positions in FIG. **1**. In other flight operations, such as immediately after a normal touchdown of an aircraft, the blocker doors **122** can be moved to their respective deployed positions to block the fluid flow FF from exiting the aft end **104**, as otherwise occurs during normal flight. In this way, the blocker doors **122** physically redirect the fluid flow FF to ultimately produce a reverse-thrust effect. The redirected fluid flow can be directed by the blocker doors **122** into cascade sets or redirecting structures to further redirect the fluid flow FF to produce a reverse-thrust effect. To move a blocker door from the stowed position to the deployed position, the trailing or aft end of the blocker door, as considered from its orientation in the stowed position, is moved radially inward (toward the central axis CA) and axially forward (toward the forward end **102**). In the deployed position, the blocker doors **122** can be arranged to be substantially perpendicular (e.g., at least within 10° of perpendicular) to the axial direction A to substantially block the fluid flow FF from exiting the aft end **104**. To move the blocker door from the deployed position to the stowed position, the trailing or aft end of the blocker door is moved radially outward (away from the central axis CA) and axially aft (toward the aft end **104**). An example acoustically-treated blocker door is provided below.

[0039] FIGS. **2** and **3** depict various views of a blocker door **200** according to an example aspect of the present disclosure. The blocker door **200** can be incorporated into the thrust reversing system **120** of the aft fan duct assembly **100** of FIG. **1**, for example. FIG. **2** is a view of a side of the blocker door **200** that faces a flow path (e.g., a bypass flow path) and FIG. **3** is a view of the blocker door **200** flipped over with respect to its orientation in FIG. **2**.

[0040] As shown, the blocker door **200** has a door body **202** and a facesheet **204** (FIGS. **4** and **5**) coupled with the door body **202**. The facesheet **204** can be mechanically fastened to the door body **202** via one or more fasteners **206** (FIG. **5**), for example. The facesheet **204** has been removed for illustrative purposes in FIG. **2** and is not visible in FIG. **3**. In at least some examples, the door body **202** can be formed from an aluminum block (e.g., the door body **202** can be machined or “hogged out”) and the facesheet **204** can be formed as a composite structure. In other example aspects, the door body **202** can be formed by other processes, such as by thermoplastic compression molding, thermoplastic injection molding, additive manufacturing, etc.

[0041] For reference, the blocker door **200** defines a first direction X, a second direction Y, and a third direction Z. The first, second, and third directions X, Y, Z are mutually perpendicular to one another and form an orthogonal direction system. The blocker door **200** also defines a midline **208** that splits the blocker door **200** into a first side **210** and a second side **212**. The midline **208** extends along the first direction X. The blocker door **200** has a forward end **214**, or hinge end, and an aft end **216**, or free end. The “forward” and “aft” ends of the blocker door **200** are designated with respect to the blocker door **200** being disposed in a stowed position. Further, the blocker door **200** has a first end **218** and a second end **220**. The door body **202** has a length extending along the first direction X, a width extending along the second direction Y, and a height extending along the third direction Z. The width of the door body **202** can be variable. In this example, the width of the door body **202** generally tapers toward the aft end **216** as shown in FIG. **2**.

[0042] The door body **202** has a backplate **222** that forms a base structure of the door body **202**. As shown in FIG. **2**, various structures extend from the backplate **222**, including various walls. In this example, an outer wall **224** extends from the backplate **222** along a portion of a perimeter of the door body **202**. The outer wall **224** generally has a U-shaped when viewed along the third direction Z, as in FIG. **2**. The outer wall **224** has a height extending along the third direction Z. The outer wall **224** has a plurality of integrated outer wall bosses **226** or threaded inserts, which are spaced from one another along the perimeter of the door body **202**. In some alternative aspects, the outer wall bosses **226** or threaded inserts of the outer wall **224** can be omitted and the facesheet **204** (FIGS. **4** and **5**) can be coupled with the door body **202** using an alternative attachment scheme, such as thermoplastic welding, adhesive bonding, etc.

[0043] A first-side partition rib **228** connects to the outer wall **224**, e.g., at the first end **218**, and extends generally along the second direction Y. A second-side partition rib **230** connects to the outer wall **224**, e.g., at the second end **220**, and extends generally along the second direction Y. The first-side partition rib **228** and the second-side partition rib **230** both connect to a mounting base **232**. The mounting base **232** has a wall **234** that defines a slot **236**, which is configured to receive a drag link (see e.g., drag links **126** extending through respective slots of the blocker doors **122** in FIG. **1**). In at least some aspects, the wall **234** has a same or similar height as the outer wall **224** and the first-side and second-side partition ribs **228**, **230**. As shown in FIG. **3**, the mounting base **232** has various mounting structures, including a first clevis **238** and a pair of second clevises **240**, **242**. The backplate **222**, the wall **234** of the mounting base **232**, and arms of the first clevis **238** define a pair of recesses **244**, **246** on opposing sides of the first clevis **238**. The recesses **244**, **246** and openings in the arms of the first clevis **238** allow for a pin or like structure to couple a drag link with the blocker door **200**. The second clevises **240**, **242** may also receive respective pins to couple the drag link with the blocker door **200**. These mounting structures can be formed integrally with the other portions of the door body **202** or can be bolted to the backplate **222**, for example.

[0044] The door body **202** can include other mounting structures in addition to the mounting structures noted above, including a first hinge **248** and a second hinge **250**. In at least some aspects, the first and second hinges **248**, **250** can be integrally formed with the backplate **222**. In other aspects, the first and second hinges **248**, **250** can be bolted or otherwise secured to the backplate **222**. Apertures of the first and second hinges **248**, **250** can receive respective pins or like structures that can pivotably couple the blocker door **200**, e.g., with an outer wall of an aft fan duct assembly. The backplate **222** defines a first hinge recess **252** associated with the first hinge **248** and a second hinge recess **254** associated with the second hinge **250**. The first hinge **248** separates the first hinge recess **252** into separate inner and outer sections **252**A, **252**B and the second hinge **250** separates the second hinge recess **254** into separate inner and outer sections **254**A, **254**B. The first and second hinge recesses **252**, **254** can reduce the height of the first and second hinges **248**, **250** with respect to the backplate **222** and can facilitate receipt of pins or like structures in the apertures of the first and second hinges **248**, **250**. In at least some examples, the outer section **252**B of the first hinge recess **252** can become gradually shallower as it (the first hinge recess **252**) extends from the first hinge **248** toward the first end **218**. Similarly, the outer section **254**B of the second hinge recess **254** becomes gradually shallower as it (the second hinge recess **254**) extends from the second hinge **250** toward the second end **220**. A flange **256** extends from the backplate **222**, e.g., at the forward end **214**. The flange **256** can extend a full width of the door body **202**.

[0045] As further shown in FIG. **2**, various structurally-active ribs or stiffeners protrude or extend from the backplate **222** to define a plurality of cavities. As illustrated in FIG. **2**, the door body **202** includes a plurality of ribs, including the first-side partition rib **228**, the second-side partition rib **230**, a central-aft rib **258**, a first-side aft rib **260**, a second-side aft rib **262**, a first-side central rib **264**, a second-side central rib **266**, a first-side central diagonal rib **268**, a second-side central diagonal rib **270**, a first-side forward diagonal rib **272**, a second-side forward diagonal rib **274**, a first-side hinge rib **276**, and a second-side hinge rib **278**. One or more of the ribs can include bosses or threaded inserts, which are arranged to receive fasteners to couple the facesheet **204** (FIGS. **4** and **5**) with the door body **202**. In at least some example aspects, the ribs can be arranged to efficiently carry loads rather than optimized to create cavities for acoustic purposes. In some aspects, the ribs, outer wall **224**, and wall **234** are at a same or complementary height with respect to one another so that the facesheet **204** (FIGS. **4** and **5**) can be overlaid onto the door body **202** flat or relatively flat. In some example aspects, the structurally-active ribs can be integrally formed with the backplate as a monolithic unitary component. In other example aspects, the structurally-active ribs can be formed and secondarily bonded or joined to the backplate **222** and/or to the facesheet **204** (FIGS. **4** and **5**). In yet other example aspects, the structurally-active ribs can be integrally formed with the facesheet **204** and then secondarily arranged relative to the door body **202** so as to define acoustic cells.

[0046] The central-aft rib **258** extends between an aft portion of the outer wall **224** and a central rib juncture **280**. The central-aft rib **258** extends parallel with the midline **208**. In the example of FIG. **2**, the central-aft rib **258** is coplanar with the midline **208**. The first-side aft rib **260** extends between a first-side aft portion of the outer wall **224** and the central rib juncture **280**. The first-side aft rib **260** is angled (e.g., by 45°) with respect to the midline **208**. In this way, the first-side aft rib **260** is arranged diagonally with respect to the first direction X. The second-side aft rib **262** generally mirrors the first-side aft rib **260** and extends between a second-side aft portion of the outer wall **224** and the central rib juncture **280**. The second-side aft rib **262** is angled with respect to the midline **208**, e.g., by a same angle as first-side aft rib **260** is angled with respect to the midline **208** but with an opposite sign. Thus, the second-side aft rib **262** is arranged diagonally with respect to the first direction X. The first-side aft rib **260** connects to the outer wall **224** at the first end **218** while the second-side aft rib **262** connects to the outer wall **224** at the second end **220**.

[0047] The first-side central rib **264** extends between a first-side central portion of the outer wall **224** and the central rib juncture **280**. The first-side central rib **264** extends generally along the second direction Y. The second-side central rib **266** mirrors the first-side central rib **264** and extends between a second-side central portion of the outer wall **224** and the central rib juncture **280**. The second-side central rib **266** extends generally along the second direction Y. The first-side central rib **264** connects to the outer wall **224** at the first end **218** while the second-side central rib **266** connects to the outer wall **224** at the second end **220**.

[0048] The first-side central diagonal rib **268** extends between the central rib juncture **280** and a juncture of a first-side forward portion of the outer wall **224** and the first-side partition rib **228**. The first-side central diagonal rib **268** supports the first-side aft portion of the mounting base **232**. The first-side central diagonal rib **268** is angled (e.g., by 45°) with respect to the midline **208**. In this way, the first-side central diagonal rib **268** is arranged diagonally with respect to the first direction X. The first-side central diagonal rib **268** extends substantially perpendicular (e.g., at least within 10° of perpendicular) to the first-side aft rib **260**. The second-side central diagonal rib **270** mirrors the first-side central diagonal rib **268** and extends between the central rib juncture **280** and a juncture of a second-side forward portion of the outer wall **224** and the second-side partition rib **230**. The second-side central diagonal rib **270** is angled with respect to the midline **208**, e.g., by a same angle as the first-side central diagonal rib **268** is angled with respect to the midline **208** but with an opposite sign. Thus, the second-side central diagonal rib **270** is arranged diagonally with respect to the first direction X. The second-side central diagonal rib **270** extends substantially perpendicular (e.g., at least within 10° of perpendicular) to the second-side aft rib **262**. The second-side central diagonal rib **270** supports the second-side aft portion of the mounting base **232**.

[0049] In the example of FIG. **2**, the central rib juncture **280** provides an intersection of seven (7) ribs. In this regard, the central rib juncture **280** provides a centralized stiffening structure to the blocker door **200** and allows loads to be distributed to the various ribs intersecting with the central rib juncture **280**. The central rib juncture **280** also provides stiffness and structural support to the mounting base **232**. The central rib juncture **280** can be coplanar, at least in part, with the midline **208**.

[0050] The first-side forward diagonal rib **272** extends between a first-side central portion of the outer wall **224** and a juncture of the first-side partition rib **228** and a forward portion of the mounting base **232**. Accordingly, the first-side forward diagonal rib **272** supports the first-side forward portion of the mounting base **232**. The first-side forward diagonal rib **272** connects to the outer wall **224** at the first end **218**. The first-side forward diagonal rib **272** is angled (e.g., by 40°) with respect to the midline **208**. In this way, the first-side forward diagonal rib **272** is arranged diagonally with respect to the first direction X. In at least one example, such as in FIG. **2**, the first-side forward diagonal rib **272** extends substantially perpendicular (e.g., at least within 10° of perpendicular) to the first-side central diagonal rib **268**. The first-side central diagonal rib **268** and the first-side forward diagonal rib **272** intersect at a first-side rib juncture **282**. The first-side rib juncture **282**, or intersection of the first-side central diagonal rib **268** and the first-side forward diagonal rib **272**, provides stiffness and structural support to the blocker door **200** at the first side **210**.

[0051] The second-side forward diagonal rib **274** mirrors the first-side forward diagonal rib **272** and extends between a second-side central portion of the outer wall **224** and a juncture of the second-side partition rib **230** and a forward portion of the mounting base **232**. Thus, the second-side forward diagonal rib **274** supports the second-side forward portion of the mounting base **232**. The second-side forward diagonal rib **274** connects to the outer wall **224** at the second end **220**. The second-side forward diagonal rib **274** is angled with respect to the midline **208**, e.g., by a same angle as the first-side forward diagonal rib **272** is angled with respect to the midline **208** but with an opposite sign. In this way, the second-side forward diagonal rib **274** is arranged diagonally with respect to the first direction X. In at least one example, such as in FIG. **2**, the second-side forward diagonal rib **274** extends substantially perpendicular (e.g., at least within 10° of perpendicular) to the second-side central diagonal rib **270**. The second-side central diagonal rib **270** and the second-side forward diagonal rib **274** intersect at a second-side rib juncture **284**. The second-side rib juncture **284**, or intersection of the second-side central diagonal rib **270** and the second-side forward diagonal rib **274**, provides stiffness and structural support to the blocker door **200** at the second side **212**. The first-side forward diagonal rib **272** and the second-side forward diagonal rib **274** converge toward one another as they extend toward the forward end **214**. Stated differently, the first-side forward diagonal rib **272** and the second-side forward diagonal rib **274** diverge away from one another as they extend toward the aft end **216**.

[0052] The first-side hinge rib **276** extends between the flange **256** and a juncture of the first-side partition rib **228**, the first-side forward diagonal rib **272**, and the mounting base **232**. The first-side hinge rib **276** extends generally along the first direction X. The second-side hinge rib **278** extends between the flange **256** and a juncture of the second-side partition rib **230**, the second-side forward diagonal rib **274**, and the mounting base **232**. The second-side hinge rib **278** extends generally along the first direction X. The first-side hinge rib **276** and the second-side hinge rib **278** are spaced from one another, e.g., along the second direction Y, and in at least some examples, are parallel to one another.

[0053] The ribs and other structures of the door body **202** can define a plurality of cavities (e.g., non-uniform cavities). For the example of FIG. **2**, the ribs and other structures of the door body **202** define a number of distinct cavities, including a first-side aft inner cavity **286** (a first cavity), a second-side aft inner cavity **288** (a second cavity), a first-side aft outer cavity **290** (a third cavity), a second-side aft outer cavity **292** (a fourth cavity), a first-side central cavity **294** (a fifth cavity), a second-side central cavity **296** (a sixth cavity), a first-side forward inner cavity **298** (a seventh cavity), a second-side forward inner cavity **300** (an eighth cavity), a first-side forward outer cavity **302** (a ninth cavity), a second-side forward outer cavity **304** (a tenth cavity), a first-side forward cavity **306** (an eleventh cavity), a second-side forward cavity **308** (a twelfth cavity), a first-side hinge cavity **310** (a thirteenth cavity), a second-side hinge cavity **312** (a fourteenth cavity), and a central forward cavity **314** (a fifteenth cavity). In at least some example aspects, each cavity is defined by at least two structurally-active ribs. In at least some example aspects, each cavity of the plurality of cavities has a length (e.g., a length extending across a given cavity) that is at least greater than ⅜ of an inch. In this regard, the blocker door **200** can be deemed a blocker door with a “large cavity acoustic structure”.

[0054] The first-side aft inner cavity **286** is defined by the central-aft rib **258**, the first-side aft rib **260**, and the outer wall **224**. In at least some aspects, the first-side aft inner cavity **286** has a generally triangular shape. The second-side aft inner cavity **288** is defined by the central-aft rib **258**, the second-side aft rib **262**, and the outer wall **224**. In at least some aspects, the second-side aft inner cavity **288** has a generally triangular shape and generally mirrors the first-side aft inner cavity **286**. The first-side aft outer cavity **290** is defined by the first-side aft rib **260**, the first-side central rib **264**, and the outer wall **224**. In at least some aspects, the first-side aft outer cavity **290** has a generally triangular shape. The second-side aft outer cavity **292** is defined by the second-side aft rib **262**, the second-side central rib **266**, and the outer wall **224**. In at least some aspects, the second-side aft outer cavity **292** has a generally triangular shape and mirrors the first-side aft outer cavity **290**.

[0055] The first-side central cavity **294** is defined by the first-side central rib **264**, the first-side central diagonal rib **268**, and the first-side forward diagonal rib **272**. In at least some aspects, the first-side central cavity **294** has a generally triangular shape. The second-side central cavity **296** is defined by the second-side central rib **266**, the second-side central diagonal rib **270**, and the second-side forward diagonal rib **274**. In at least some aspects, the second-side central cavity **296** has a generally triangular shape and generally mirrors the first-side central cavity **294**.

[0056] The first-side forward inner cavity **298** is defined by the first-side central diagonal rib **268**, the first-side forward diagonal rib **272**, and the mounting base **232**. In at least some aspects, the first-side forward inner cavity **298** has a generally triangular shape. However, the mounting base **232** has a stepped sidewall **316** that steps the mounting base **232** inward toward the midline **208**, which increases the volume of the first-side forward inner cavity **298**. The second-side forward inner cavity **300** is defined by the second-side central diagonal rib **270**, the second-side forward diagonal rib **274**, and the mounting base **232**. In at least some aspects, the second-side forward inner cavity **300** has a generally triangular shape and mirrors the first-side forward inner cavity **298**. However, the mounting base **232** has a stepped sidewall **318** that steps the mounting base **232** inward toward the midline **208**, which increases the volume of the second-side forward inner cavity **300**.

[0057] The first-side forward outer cavity **302** is defined by the first-side central diagonal rib **268**, the first-side forward diagonal rib **272**, and the outer wall **224**. In at least some aspects, the first-side forward outer cavity **302** has a generally triangular shape. The second-side forward outer cavity **304** is defined by the second-side central diagonal rib **270**, the second-side forward diagonal rib **274**, and the outer wall **224**. In at least some aspects, the second-side forward outer cavity **304** has a generally triangular shape and mirrors the first-side forward outer cavity **302**.

[0058] The first-side forward cavity **306** is defined by the first-side central diagonal rib **268**, the first-side forward diagonal rib **272**, and the first-side partition rib **228**. In at least some aspects, the first-side forward cavity **306** has a generally triangular shape. The second-side forward cavity **308** is defined by the second-side central diagonal rib **270**, the second-side forward diagonal rib **274**, and the second-side partition rib **230**. In at least some aspects, the second-side forward cavity **308** has a generally triangular shape and mirrors the first-side forward cavity **306**.

[0059] The first-side hinge cavity **310** is defined by the outer wall **224**, the first-side partition rib **228**, the flange **256**, and the first-side hinge rib **276**. The first-side hinge cavity **310** extends the length of the first hinge recess **252** along the first direction X. For this example, the first-side hinge cavity **310** gradually increases in depth from the first hinge **248** (FIG. **3**) to the first end **218**, and inversely, the outer section **252**B of the first hinge recess **252** can become gradually shallower, as noted previously. Accordingly, the first-side hinge cavity **310** mirrors the first hinge recess **252** along a thickness direction of the blocker door **200**, or rather, the third direction Z. The depth of the first-side hinge cavity **310** can remain constant from the first hinge **248** to the first-side hinge rib **276**. In this regard, the first-side hinge cavity **310** has a constant depth portion (e.g., between the first hinge **248** and the first-side hinge rib **276**) and a variable depth portion (e.g., between the first hinge **248** and the first end **218**). The ramped or variable depth portion of the first-side hinge cavity **310** can advantageously increase the volume of the first-side hinge cavity **310**, which can enhance the acoustic attenuation of the blocker door **200**.

[0060] The second-side hinge cavity **312** mirrors the first-side hinge cavity **310** and is defined by the outer wall **224**, the second-side partition rib **230**, the flange **256**, and the second-side hinge rib **278**. The second-side hinge cavity **312** extends the length of the second hinge recess **254** along the first direction X. For this example, the second-side hinge cavity **312** gradually increases in depth from the second hinge **250** (FIG. **3**) to the second end **220**, and inversely, the outer section **254**B of the second hinge recess **254** can become gradually shallower, as noted previously. Accordingly, the second-side hinge cavity **312** mirrors the second hinge recess **254** along a thickness direction of the blocker door **200**, or rather, the third direction Z. The depth of the second-side hinge cavity **312** can remain constant from the second hinge **250** to the second-side hinge rib **278**. In this way, the second-side hinge cavity **312** has a constant depth portion (e.g., between the second hinge **250** and the second-side hinge rib **278**) and a variable depth portion (e.g., between the second hinge **250** and the second end **220**). The ramped or variable depth portion of the second-side hinge cavity **312** can advantageously increase the volume of the second-side hinge cavity **312**, which can enhance the acoustic attenuation of the blocker door **200**.

[0061] The central forward cavity **314** is generally defined between the first-side hinge cavity **310** and the second-side hinge cavity **312**, e.g., along the first direction X, and is defined between the flange **256** and the mounting base **232**, e.g., along the second direction Y. Accordingly, the central forward cavity **314** is defined by the mounting base **232**, the flange **256**, the first-side hinge rib **276**, and the second-side hinge rib **278**.

[0062] While FIG. **2** depicts one example arrangement in which the structurally-active ribs can be arranged to define the cavities, the structurally-active ribs can have other arrangements in other example aspects. For instance, in other example aspects, more or less ribs can be included, the ribs can be arranged asymmetrically with respect to the midline **208**, the ribs can have other symmetric arrangements than the one illustrated in FIG. **2**, etc.

[0063] With reference now to FIGS. **4** and **5**, FIG. **4** shows an exploded view of the blocker door **200** and FIG. **5** shows an assembled version of the blocker door **200**. As shown, the facesheet **204** can be laid over the door body **202** and coupled thereto. The facesheet **204** is generally shaped to fit or overlay over an entirety of the door body **202**, e.g., as viewed along the third direction Z. The facesheet **204** contacts or is seated on the flange **256**, top surfaces of the structurally-active ribs, the top surface of the outer wall **224**, and the wall **234** of the mounting base **232**. The facesheet **204** can define a plurality of apertures **320** that can be aligned in communication with the bosses or threaded inserts of the door body **202** and, if applicable, openings defined in the flange **256** (see FIG. **4**). The apertures **320** can be circular openings arranged to receive fasteners therethrough, for example. Accordingly, in at least some examples, the facesheet **204** can be mechanically fastened to the door body **202** via fasteners **206**. In other examples, the facesheet **204** can be bonded using thermoplastic welding, adhesive bonding, etc.

[0064] The facesheet **204** can also define a plurality of perforations **322**, such as the slotted perforations shown in FIGS. **4** and **5**. The perforations **322** extend lengthwise along the second direction Y in this example. In other example aspects, the perforations can have other shapes, such as circular openings. The facesheet **204** can define the perforations **322** so that each one of the cavities defined by the door body **202** has at least one perforation in communication therewith. In this way, with the facesheet **204** coupled with the door body **202**, the facesheet **204** encloses the plurality of cavities **286**, **288**, **290**, **292**, **294**, **296**, **298**, **300**, **302**, **304**, **306**, **308**, **310**, **312**, **314** (FIG. **2**) and the perforations **322** communicate with the plurality of cavities **286**, **288**, **290**, **292**,

294, **296**, **298**, **300**, **302**, **304**, **306**, **308**, **310**, **312**, **314** to create a plurality of acoustic cells, collectively acoustic cells **324**. In this regard, the acoustic cells **324** are created by their respective cavities being enclosed by the facesheet **204** and flow communication being provided by the perforations **322** to allow fluid (e.g., high bypass air) to flow into and out of the acoustic cells **324**. The acoustic cells **324** function to attenuate noise, and more specifically, aft-radiating noise from an aircraft engine assembly.

[0065] In FIG. **5**, the acoustic cells **324** are depicted by the dashed-lined sections. In this example, the blocker door **200** includes fifteen (15) distinct acoustic cells **324**. As shown in FIG. **5**, the acoustic cells **324** include a first-side aft inner cell **326** (a first acoustic cell), a second-side aft inner acoustic cell **328** (a second acoustic cell), a first-side aft outer cell **330** (a third acoustic cell), a second-side aft outer cell **332** (a fourth acoustic cell), a first-side central cell **334** (a fifth acoustic cell), a second-side central cell **336** (a sixth acoustic cell), a first-side forward inner cell **338** (a seventh acoustic cell), a second-side forward inner cell **340** (an eighth acoustic cell), a first-side forward outer cell **342** (a ninth acoustic cell), a second-side forward outer cell **344** (a tenth acoustic cell), a first-side forward cell **346** (an eleventh acoustic cell), a second-side forward cell **348** (a twelfth acoustic cell), a first-side hinge cell **350** (a thirteenth acoustic cell), a second-side hinge cell **352** (a fourteenth acoustic cell), and a central forward cell **354** (a fifteenth acoustic cell). As illustrated, each one of the acoustic cells **324** corresponds to one of the cavities **286**, **288**, **290**, **292**, **294**, **296**, **298**, **300**, **302**, **304**, **306**, **308**, **310**, **312**, **314** (FIG. **2**) and has a plurality of perforations **322** associated therewith, which allows for flow communication between the acoustic cells **324** and e.g., the flow passage in which the blocker door **200** is positioned.

[0066] In other examples, the blocker door **200** can include more or less than fifteen (15) acoustic cells. For instance, in some example aspects, the plurality of acoustic cells **324** can range between ten to sixteen acoustic cells (including the endpoints), which can advantageously balance the structural, acoustic, and weight considerations of the blocker door **200**. In some example aspects, an average volume of the acoustic cells **324** can range between 7 and 9 cubic inches, which can advantageously balance the structural, acoustic, and weight considerations of the blocker door **200**. In yet other example aspects, the acoustic cells **324** can cover an area of the blocker door **200** that is greater than eighty percent (80%) of a total area of the blocker door **200**, e.g., as viewed along the third direction Z. In some additional example aspects, the mounting base **232** can be surrounded by acoustic cells **324**, e.g., as shown in FIG. **5**, which can make efficient use of the space of the blocker door **200**.

[0067] In some example aspects, at least one of the plurality of acoustic cells **324** is empty to allow for a volume of air to fill therein. That is, the cavities **286**, **288**, **290**, **292**, **294**, **296**, **298**, **300**, **302**, **304**, **306**, **308**, **310**, **312**, **314** can be empty cavities. In other example aspects, at least one of the plurality of acoustic cells **324** can have a volume of acoustic attenuating material disposed therein, such as a honeycomb structure.

[0068] In some alternative aspects, as noted previously, at least some of the structurally-active ribs can be formed and secondarily bonded or joined to the backplate **222** and/or to the facesheet **204**. In yet other example aspects, at least some of the structurally-active ribs can be integrally formed with the facesheet **204**, and the facesheet **204** having the structurally-active ribs can be arranged and secured relative to a door body so as to define the acoustic cells. An example is provided below.

[0069] FIG. **6** is a perspective view of a side of a facesheet **204** that confronts or faces a door body when coupled thereto, with the facesheet **204** having the plurality of structurally-active ribs integrally formed therewith or coupled thereto. As shown, the facesheet **204** can include structurally-active ribs **360** and a perimeter wall **356** that is spaced from an edge **358** of the facesheet **204**. The structurally-active ribs **360** and the perimeter wall **356** can define a plurality of cavities. When the facesheet **204** is coupled with a door body, an outer wall (e.g., the outer wall **224** in FIG. **2**) of the door body can be arranged in the space between the perimeter wall **356** and the edge **358**. In this way, the perimeter wall **356** can be positioned adjacent to the outer wall, and in some aspects, can be connected thereto, e.g., by fasteners, an adhesive, a weld, etc. A mounting base of the door body (e.g., the mounting base **232**) can be arranged within a void **362** defined by the facesheet **204**. The structurally-active ribs **360**, the perimeter wall **356**, and/or other surfaces of the facesheet **204** can engage the door body to be coupled thereto. Specifically, when the facesheet **204** is coupled with a door body, the plurality of cavities defined by the structurally-active ribs **360** arranged between the backplate of the door body and the facesheet **204** communicate with the plurality of perforations **322** to create a plurality of acoustic cells, which as noted herein, can advantageously attenuate noise.

[0070] In some alternative aspects, the outer wall of the door body to which the facesheet **204** of FIG. **6** is coupled can be omitted. In such aspects, the perimeter wall **356** of the facesheet **204** of FIG. **6** can be arranged at the edge **358** around the perimeter, and accordingly, can act as the outer wall.

[0071] FIG. **7** is a flow diagram for a method **400** of fabricating a blocker door according to an example aspect of the present disclosure.

[0072] At **402**, the method **400** can include creating a door body from a block of material, the door body having a backplate and a plurality of ribs that extend from the backplate to define a plurality of cavities. For instance, a door body can be created by forming or machining (e.g., "hogging out") an aluminum block of material. The aluminum block can be machined so that the door body has a backplate and a plurality of structurally-active ribs or stiffeners extending from the backplate to define a plurality of cavities. The aluminum block can further be machined so that the door body has an outer wall extending from the backplate and at least in part around a perimeter of the door body. The outer wall can be thicker than the ribs. The aluminum block can also be machined so that the door body has a plurality of mounting structures, such as hinges and clevises, which may enable the blocker door to be coupled within a thrust reversing system of an aft duct assembly for an aircraft engine assembly. The aluminum block can be machined so that the ribs are optimally arranged to carry loads but also so that the cavities formed thereby are intelligently arranged to provide acoustic attenuation. The ribs, outer wall, and a mounting base also machined from the aluminum block can each be at a height to allow for a facesheet to be laid flat or relatively flat thereon. In at least some aspects, the aluminum block is machined so that the ribs (and in some aspects the outer wall, mounting features, mounting base, flange, etc.) are integrally formed with the backplate as a monolithic unitary component. In some alternative aspects, the door body can be additively manufactured (e.g., 3D printed) as a monolithic unitary component, or made via other materials and processes (e.g., thermoplastic forming or thermoplastic compression molding, thermoplastic injection molding, composite layup, etc.).

[0073] At **404**, the method **400** can include coupling a facesheet to the door body to enclose the plurality of cavities and so that a plurality of perforations defined by the facesheet communicate with the plurality of cavities to create a plurality of acoustic cells. For instance, a facesheet can be laid over the door body and coupled thereto, e.g., by mechanical fasteners, adhesives, a combination thereof, etc. The facesheet can be shaped to fit or overlay over an entirety of the door body. The facesheet can define a plurality of perforations, such as the slotted perforations. The facesheet can be aligned with and coupled to the door body so that the facesheet encloses the cavities. The perforations can be arranged on the facesheet so that the perforations communicate with the cavities. The enclosed cavities having flow communication by way of the perforations creates the acoustic cells. The perforations allow fluid (e.g., high bypass air) to flow into and out of the acoustic cells, which provides noise attenuation. The acoustic cells of the blocker door can attenuate aft-radiating noise from an aircraft engine assembly, for example. In some implementations, one or more of the acoustic cells can be empty to allow for a volume of air to fill therein. In some implementations, one or more of the acoustic cells can include a volume of acoustic attenuating material disposed therein.

[0074] To summarize, the present disclosure provides a blocker door having multi-functional integral ribs or stiffeners that react flight/ground loads and also act to form resonance chambers for acoustic attenuation, which are closed with a perforated facesheet. In some aspects, a door body of the blocker door can be formed from one solid "chunk" of material (e.g., aluminum) with the ribs being integrally formed with a backplate as a monolithic unitary component. A carbon-fiber-reinforced polymer (CFRP) facesheet with perforations can enclose the cavities formed by the integrally formed ribs to create acoustic cells. The perforated facesheet can face into an aft fan duct flow region to attenuate aft-radiating fan noise. In other aspects, the door body can be formed of other materials, such as by a thermoplastic, as well as by other fabrication processes, such as by additive manufacturing, to obtain the desired, tailored cell volume to offset aft fan duct noise.
[0075] In the current disclosure, reference is made to various aspects. However, it should be understood that the present disclosure is not limited to specific described aspects. Instead, any combination of the following features and elements, whether related to different aspects or not, is contemplated to implement and practice the teachings provided herein. Additionally, when elements of the aspects are described in the form of "at least one of A and B," it will be understood that aspects including element A exclusively, including element B exclusively, and including element A and B are each contemplated. Furthermore, although some aspects may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given aspect is not limiting of the present disclosure. Thus, the aspects, features, aspects and advantages disclosed herein are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s).
[0076] While the foregoing is directed to aspects of the present disclosure, other and further aspects of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A blocker door, comprising: a door body having a backplate; a facesheet coupled with the door body and defining a plurality of perforations; and a plurality of ribs arranged between the backplate and the facesheet to define a plurality of cavities that communicate with the plurality of perforations to create a plurality of acoustic cells.
2. The blocker door of claim 1, wherein each cavity of the plurality of cavities has a length that is at least greater than ⅜ of an inch.
3. The blocker door of claim 1, wherein the blocker door defines a midline that separates the blocker door into a first side and a second side, and wherein the plurality of ribs includes a central-aft rib extending along the midline, a first-side aft rib angled with respect to the midline, a second-side aft rib angled with respect to the midline, a first-side central rib extending perpendicular to the midline, a second-side central rib extending perpendicular to the midline, a first-side central diagonal rib angled diagonal to the midline, and a second-side central diagonal rib angled diagonal to the midline, and wherein the central-aft rib, the first-side aft rib, the second-side aft rib, the first-side central rib, the second-side central rib, the first-side central diagonal rib, and the second-side central diagonal rib intersect at a central rib juncture.
4. The blocker door of claim 3, wherein the door body has an outer wall that extends from the backplate and along at least a portion of a perimeter of the door body, and wherein the central-aft rib, the first-side aft rib, the second-side aft rib, the first-side central rib, the second-side central rib, the first-side central diagonal rib, and the second-side central diagonal rib each respectively connect to the outer wall.
5. The blocker door of claim 1, wherein the door body has a hinge extending from the backplate in a direction opposite the plurality of ribs, and wherein the backplate defines a hinge recess in which the hinge is positioned, and wherein the door body includes a flange extending from the backplate,

a mounting base, an outer wall, and wherein the plurality of ribs includes a partition rib and a hinge rib that extends between and connects the flange and the mounting base, and wherein the flange, the partition rib, the hinge rib, and the outer wall define a hinge cavity, which is one of the plurality of cavities, and wherein the hinge cavity mirrors the hinge recess along a thickness direction of the blocker door.

6. The blocker door of claim 5, wherein the hinge cavity has a portion that gradually increases in depth as the hinge cavity extends toward an end of the blocker door.

7. The blocker door of claim 1, wherein the door body includes a flange extending from the backplate and a mounting base that has one or more mounting structures, and wherein the plurality of ribs includes a first-side hinge rib and a second-side hinge rib that each extend between and connect the flange and the mounting base, and wherein the plurality of cavities includes a central forward cavity defined by the flange, the mounting base, the first-side hinge rib, and the second-side hinge rib.

8. The blocker door of claim 1, wherein the door body has an outer wall that extends from the backplate and along at least a portion of a perimeter of the door body, and wherein the door body includes a mounting base that includes one or more mounting structures, and wherein the plurality of ribs includes a central diagonal rib and a forward diagonal rib that each extend at an angle with respect to a midline of the blocker door and each extend perpendicular to one another, the central diagonal rib and the forward diagonal rib intersect at a rib juncture and each extend between and connect to the outer wall and the mounting base.

9. The blocker door of claim 1, wherein the blocker door has a forward end and an aft end and defines a midline that separates the blocker door into a first side and a second side, wherein the door body has an outer wall that extends from the backplate and along at least a portion of a perimeter of the door body, and wherein the door body includes a mounting base that includes one or more mounting structures, and wherein the plurality of ribs includes a first-side forward diagonal rib and a second-side forward diagonal rib, the first-side forward diagonal rib is arranged on the first side and extends between and connects to the outer wall and the mounting base, the second-side forward diagonal rib is arranged on the second side and extends between and connects to the outer wall and the mounting base, the first-side forward diagonal rib and the second-side forward diagonal rib converge toward one another as they extend toward the forward end.

10. The blocker door of claim 1, wherein the door body includes a mounting base that includes one or more mounting structures, and wherein the plurality of cavities includes a forward inner cavity defined by the mounting base, and a central diagonal rib and a forward diagonal rib of the plurality of ribs, and wherein the mounting base has a stepped sidewall that steps the mounting base inward toward a midline defined by the blocker door.

11. The blocker door of claim 1, wherein the blocker door defines a midline separating the blocker door into a first side and a second side, and wherein the plurality of ribs are arranged so that: at least two ribs of the plurality of ribs intersect at a first-side rib juncture on the first side of the blocker door, at least at least two ribs of the plurality of ribs intersect at a second-side rib juncture on the second side of the blocker door, and at least two ribs of the plurality of ribs intersect at a central rib juncture arranged coplanar with the midline.

12. The blocker door of claim 1, wherein an average volume of the acoustic cells ranges between 7 and 9 cubic inches.

13. The blocker door of claim 1, wherein the plurality of acoustic cells include ten to sixteen acoustic cells.

14. The blocker door of claim 1, wherein at least one of the plurality of acoustic cells is empty to allow for a volume of air to fill therein.

15. The blocker door of claim 1, wherein at least one of the plurality of acoustic cells has a volume of acoustic attenuating material disposed therein.

16. The blocker door of claim 1, wherein the ribs are integrally formed with the backplate as a

monolithic unitary component.
17. An aft fan duct assembly, comprising: an inner duct wall; an outer duct wall, the inner duct wall and the outer duct wall defining a fan duct; blocker doors arranged within the fan duct and each movable between a stowed position and a deployed position, and wherein each one of the blocker doors comprises: a door body having a backplate and a plurality of ribs that extend from the backplate to define a plurality of cavities; and a facesheet coupled with the door body, wherein the facesheet encloses the plurality of cavities and defines a plurality of perforations that communicate with the plurality of cavities to create a plurality of acoustic cells.
18. The aft fan duct assembly of claim 17, wherein the plurality of ribs includes a set of ribs that intersect at a central rib juncture, wherein a first rib of the set extends along a midline defined by the blocker door, a second rib of the set extends perpendicular to the midline, and a third rib of the set extends diagonally between the first and second rib.
19. The aft fan duct assembly of claim 17, wherein the blocker door defines a midline separating the blocker door into a first side and a second side, and wherein the plurality of ribs are arranged so that: at least two ribs of the plurality of ribs intersect at a first-side rib juncture on the first side of the blocker door, at least at least two ribs of the plurality of ribs intersect at a second-side rib juncture on the second side of the blocker door, and at least two ribs of the plurality of ribs intersect at a central rib juncture arranged coplanar with the midline.
20. A method of fabricating a blocker door, comprising: creating a door body from a block of material, the door body having a backplate and a plurality of ribs that extend from the backplate to define a plurality of cavities; and coupling a facesheet to the door body to enclose the plurality of cavities and so that a plurality of perforations defined by the facesheet communicate with the plurality of cavities to create a plurality of acoustic cells.
